# AI-Assisted Decision Support Systems Using Large Language Models: A Comparative Evaluation of Prompt Engineering Strategies for Enterprise Business Analytics

*Author: Trisha Dasari*
*Master of Arts in Information Technology Management*
*Webster University – St Louis, MO, USA*
*td8m28@gmail.com*

**ABSTRACT**

**Large Language Models (LLMs) have emerged as powerful tools for enhancing enterprise decision-making by generating context-aware insights from structured and unstructured data. However, the quality, reliability, and usefulness of AI-generated recommendations depend significantly on the prompting strategy used during interaction with these models. This research investigates the application of prompt engineering techniques for AI-assisted decision support in enterprise business analytics. A comparative study was conducted using the Tableau Sample Superstore dataset containing 9,994 retail transactions and 21 business attributes. Python-based exploratory data analysis (EDA) was performed using Pandas, NumPy, and Matplotlib to generate descriptive business summaries, including sales performance, profitability, regional distribution, product categories, and customer segments.**

**The generated business summaries were provided to Google Gemini 2.5 Flash through four distinct prompting strategies: Zero-Shot Prompting, Structured Prompting, Role-Based Prompting, and Few-Shot Prompting. The responses were comparatively evaluated using qualitative and quantitative criteria, including business understanding, numerical accuracy, evidence-based reasoning, hallucination resistance, handling of incomplete information, executive readability, and decision quality.**

**Experimental results demonstrate that prompting strategy has a significant influence on the quality of AI-generated business recommendations. Structured and Few-Shot prompting consistently produced the most accurate, transparent, and evidence-based outputs, while Role-Based prompting generated highly actionable recommendations suitable for executive decision-making. Zero-Shot prompting provided comprehensive insights but occasionally introduced unsupported assumptions beyond the available evidence. The findings highlight the importance of prompt engineering as a practical approach for improving the reliability and trustworthiness of AI-assisted enterprise decision support systems. This study contributes a reproducible evaluation framework that can assist researchers and organizations in selecting effective prompting strategies for business analytics applications and provides a foundation for future research involving multi-model comparisons, Retrieval-Augmented Generation (RAG), and Explainable Artificial Intelligence (XAI).**

***Keywords -*** *Artificial Intelligence, Large Language Models, Prompt Engineering, Enterprise Decision Support Systems, Business Analytics, Google Gemini, Explainable AI, Responsible AI, Prompt Evaluation.*

## I. INTRODUCTION

Artificial Intelligence (AI) has become a transformative technology across modern enterprises, enabling organizations to improve operational efficiency, automate repetitive tasks, and support strategic decision-making through data-driven insights. With the rapid growth of digital business ecosystems, organizations generate vast amounts of structured and unstructured data every day. Extracting meaningful knowledge from this data has become increasingly challenging, creating a growing demand for intelligent decision support systems capable of assisting business leaders in making timely and informed decisions.

Decision Support Systems (DSS) have traditionally combined analytical models, statistical techniques, and historical business data to assist managers in evaluating business performance and identifying strategic opportunities. Recent advances in Large Language Models (LLMs), including Google's Gemini, OpenAI's GPT models, Anthropic's Claude, and Meta's Llama, have significantly expanded the capabilities of modern decision support systems. Unlike traditional analytics tools, LLMs can interpret natural language instructions, summarize complex datasets, generate contextual business insights, and recommend strategic actions in a human-readable format. These capabilities have accelerated the adoption of AI-powered enterprise analytics across industries such as finance, healthcare, retail, manufacturing, logistics, and customer relationship management.

Despite these advancements, the effectiveness of AI-generated recommendations depends heavily on the quality of user interaction with the model. Large Language Models are highly sensitive to prompt design, and poorly structured prompts may lead to incomplete reasoning, inconsistent outputs, or unsupported conclusions, commonly referred to as hallucinations. Consequently, prompt engineering has emerged as a critical technique for improving the reliability, transparency, and factual consistency of AI-generated responses. Different prompting strategies including Zero-Shot Prompting, Structured Prompting, Role-Based Prompting, and Few-Shot Prompting provide varying levels of contextual guidance to the model, influencing its reasoning process and the overall quality of generated recommendations.

In enterprise environments, where business decisions often involve financial investment, operational planning, and risk management, the accuracy and trustworthiness of AI-generated recommendations are of paramount importance. Organizations require AI systems that not only generate insightful recommendations but also clearly distinguish evidence-based conclusions from assumptions while appropriately handling incomplete or uncertain information. Therefore, understanding how different prompting strategies influence the quality of AI-assisted decision support has become an important area of research.

This study investigates the application of prompt engineering techniques for AI-assisted enterprise decision support using a real-world retail sales dataset. Business performance data obtained from the Tableau Sample Superstore dataset were analyzed using Python-based exploration data analysis to generate structured business summaries. These summaries were subsequently evaluated using Google Gemini 2.5 Flash under four prompting strategies: Zero-Shot Prompting, Structured Prompting, Role-Based Prompting, and Few-Shot Prompting. The responses generated by each strategy were systematically compared using qualitative and quantitative evaluation criteria, including business understanding, numerical accuracy, evidence-based reasoning, hallucination resistance, handling of incomplete information, executive readability, and decision quality.

The primary objectives of this research are to

- ✓ Evaluate the influence of different prompting strategies on AI-generated business recommendations,
- ✓ Compare the reliability and consistency of multiple prompt engineering approaches
- ✓ Identify prompting techniques that minimize unsupported reasoning while improving evidence-based decision support
- ✓ Propose a reproducible evaluation framework that can assist researchers and organizations in developing trustworthy AI-assisted enterprise decision support systems.

The findings of this research contribute to the growing field of prompt engineering by demonstrating how carefully designed prompts can substantially improve the effectiveness, transparency, and practical applicability of Large Language Models in enterprise business analytics.

## II. LITERATURE REVIEW

Artificial Intelligence (AI) has significantly transformed enterprise information systems by enabling organizations to extract actionable insights from large volumes of structured and unstructured data. Enterprise Decision Support Systems (DSS), Large Language Models (LLMs), Prompt Engineering, Explainable Artificial Intelligence (XAI), Responsible AI, and Business Analytics have become key research areas for developing intelligent decision-support solutions. This section reviews the existing literature related to these domains and highlights the research gap addressed in this study.

### A. Enterprise Decision Support Systems

Decision Support Systems (DSS) are computer-based systems that assist managers in solving semi-structured and unstructured business problems through data analysis, modeling, and visualization. Traditional DSS integrate databases, statistical techniques, optimization models, and reporting tools to support strategic planning and operational decision-making. Turban et al. [11] describe DSS as an integration of data management, analytical models, and user interfaces that enhance organizational decision quality. Similarly, Alter [12] emphasized that DSS improve managerial effectiveness by supporting interactive problem-solving rather than replacing human judgment.

With the rapid growth of enterprise data, Business Intelligence (BI) platforms have evolved from descriptive reporting tools to AI-assisted decision-support environments. Davenport and Harris [10] demonstrated that organizations adopting data-driven decision-making and advanced analytics consistently outperform competitors in operational efficiency and strategic planning. These developments have encouraged the integration of Artificial Intelligence into enterprise analytics, enabling organizations to automate reporting, identify hidden business patterns, and improve decision quality.

### B. Large Language Models for Enterprise Analytics

The introduction of Transformer architecture revolutionized Natural Language Processing (NLP) by enabling models to process long-range contextual information efficiently. Vaswani et al. [1] introduced Transformer architecture, which eliminated recurrent neural networks and significantly improved sequence modeling through self-attention mechanisms. This architecture became the foundation for modern Large Language Models.

Building on the Transformer architecture, Brown et al. [2] introduced GPT-3, demonstrating that LLMs possess strong few-shot learning capabilities without requiring task-specific fine-tuning. Subsequent research has expanded these capabilities to reasoning, summarization, question answering, and enterprise knowledge management. The GPT-4 Technical Report further demonstrated improved reasoning, instruction-following, and multimodal capabilities while emphasizing the importance of safety evaluations and responsible deployment [4]. Similarly, Google's Gemini model family introduced advanced multimodal reasoning capabilities designed for enterprise and research applications [5].

These developments have significantly increased the adoption of LLMs in enterprise analytics, where organizations use AI systems to summarize business reports, generate strategic recommendations, automate documentation, and support executive decision-making. However, research consistently indicates that LLM output remains highly dependent on prompt quality and may generate unsupported conclusions when insufficient contextual guidance is provided [4], [5].

### C. Prompt Engineering

Prompt engineering has emerged as one of the most important techniques for improving the reliability and performance of Large Language Models. Instead of modifying model parameters, prompt engineering focuses on designing effective instructions that guide model reasoning and output generation.

Brown et al. [2] demonstrated the effectiveness of Zero-Shot and Few-Shot prompting for adapting language models to unseen tasks. Wei et al. [3] further introduced Chain-of-Thought prompting, showing that structured reasoning significantly improves performance on complex reasoning tasks. More recent systematic reviews have shown that prompt engineering has evolved into a structured research area with reusable prompt patterns, taxonomies, and evaluation frameworks. Sasaki et al. reviewed prompt engineering patterns in software engineering and proposed a taxonomy that categorizes prompt strategies based on their practical applications. Their study highlights the importance of structured prompts in improving consistency and reducing ambiguity in LLM outputs. (Waseda University)

Similarly, Huang et al. conducted a systematic literature review of prompt engineering and identified major research gaps related to reproducibility, prompt evaluation, and trustworthy deployment of LLM-based systems. Their work emphasizes that effective prompt engineering remains one of the primary challenges for achieving reliable AI-assisted decision support. (Monash University)

The present study builds upon these prompting techniques by comparatively evaluating Zero-Shot, Structured, Role-Based, and Few-Shot prompting within an enterprise business analytics scenario.

### D. Explainable Artificial Intelligence

As AI systems become increasingly integrated into business processes, transparency and interpretability have become essential requirements for organizational trust. Explainable Artificial Intelligence (XAI) seeks to improve user confidence by providing understandable explanations for AI-generated predictions and recommendations.

Ribeiro et al. introduced the Local Interpretable Model-Agnostic Explanations (LIME) framework, enabling users to understand predictions generated by complex machine learning models [7]. Similarly, Lundberg and Lee proposed SHAP (SHapley Additive exPlanations), providing a theoretically grounded approach for interpreting feature contributions in predictive models [6].

Although LLMs differ from traditional supervised learning models, explainability principles remain relevant. Prompt engineering can encourage models to produce transparent reasoning, explicitly acknowledge uncertainty, and distinguish evidence-based conclusions from assumptions, thereby improving user trust in AI-assisted decision support.

### E. Responsible Artificial Intelligence

The increasing adoption of LLMs in enterprise environments has raised important concerns regarding transparency, fairness, accountability, robustness, and reliability. Responsible AI frameworks aim to ensure that AI systems operate safely while supporting ethical decision-making.

The National Institute of Standards and Technology (NIST) introduced the AI Risk Management Framework (AI RMF 1.0), providing guidance for developing trustworthy AI systems through governance, risk assessment, transparency, and continuous monitoring [8]. NIST later extended this work with a Generative AI Profile that specifically addresses risks associated with foundation models and generative AI systems [9]. Microsoft's Responsible AI Standard similarly outlines principles for fairness, reliability, privacy, inclusiveness, transparency, and accountability in AI deployment [13].

These frameworks collectively emphasize that enterprise AI systems should generate recommendations supported by available evidence, clearly communicate uncertainty, and avoid fabricated information. Such principles directly motivate the evaluation criteria adopted in this research.

### F. AI-Assisted Business Analytics

Business Analytics combines statistical analysis, machine learning, visualization, and predictive modeling to transform organizational data into actionable business knowledge. Advances in LLMs have enabled natural-language interfaces that simplify analytical workflows and make business intelligence more accessible to non-technical users.

Despite rapid progress, most existing studies focus on benchmarking language understanding tasks rather than evaluating prompt engineering for practical enterprise decision support. Recent work exploring prompt engineering for decision-making argues that carefully designed prompts improve reasoning quality and help align LLM outputs with business objectives.

Therefore, there remains a need for reproducible evaluation frameworks that compare prompting strategies using real-world business datasets. This research addresses that need by integrating Python-based exploratory data analysis with systematic prompt evaluation using Google Gemini to compare Zero-Shot, Structured, Role-Based, and Few-Shot prompting. The study contributes empirical evidence on how prompt design influences the quality, transparency, and reliability of AI-generated business recommendations for enterprise decision support.

## III. PROPOSED WORK

The rapid advancement of Large Language Models (LLMs) has created new opportunities for integrating Artificial Intelligence into enterprise decision-support systems. Despite their remarkable reasoning capabilities, the quality of AI-generated recommendations is highly dependent on the prompting strategy used to communicate analytical tasks. Different prompt designs may lead to variations in factual accuracy, reasoning quality, consistency, and transparency, making prompt engineering a critical component of trustworthy AI-assisted decision support.
The primary objective of this research is to develop and evaluate an AI-assisted enterprise decision-support framework that combines traditional data analytics with Large Language Models to generate evidence-based business recommendations. Unlike conventional Business Intelligence (BI) systems that primarily focus on descriptive reporting and visualization, the proposed framework integrates Python-based data analysis with prompt engineering techniques to produce contextual and actionable business insights using an LLM.

The proposed framework begins with the **Tableau Sample Superstore dataset**, which contains **9,994 retail transactions** and **21 business attributes**, including sales, profit, customer segments, product categories, shipping information, and geographic regions. Exploratory Data Analysis (EDA) is performed using **Python**, employing the **Pandas**, **NumPy**, and **Matplotlib** libraries to clean the dataset, generate descriptive statistics, identify business trends, and visualize regional sales performance.

Following the exploratory analysis, a structured **Business Summary** is automatically generated from the analytical results. This summary contains key business metrics, including total sales, total profit, regional performance, top-performing product categories, customer segment analysis, and geographical sales distribution. The generated business summary serves as the standardized input for all prompting experiments, ensuring that each Large Language Model receives identical business information throughout the evaluation process.

The business summary is subsequently provided to **Google Gemini Flash** using four different prompt engineering strategies:

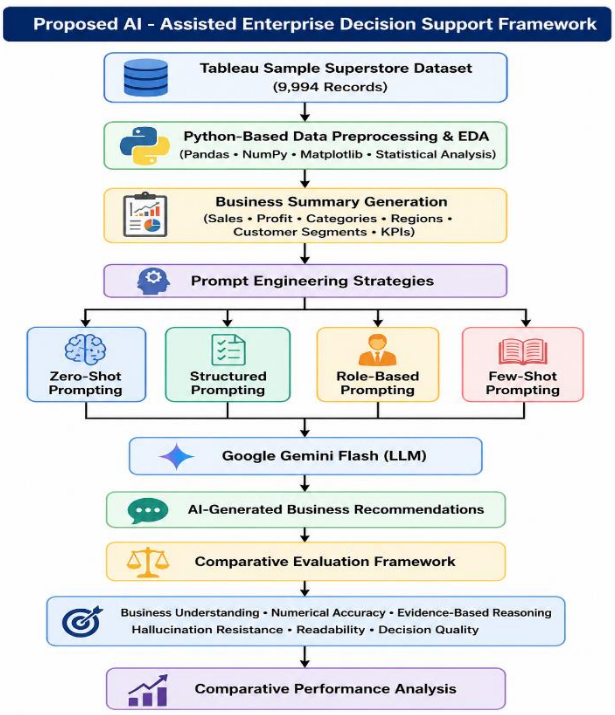
- **Zero-Shot Prompting**, where the model receives only the analytical task without any examples.
- **Structured Prompting**, which guides the model through a predefined reasoning sequence to encourage evidence-based analysis.
- **Role-Based Prompting**, where the model assumes the role of a Chief Strategy Officer responsible for executive decision-making.
- **Few-Shot Prompting**, where the model receives an example analysis before performing the business evaluation.

Each prompting strategy is evaluated independently while maintaining identical business data to ensure a fair comparison.
The responses generated by the LLM are then assessed using a comparative evaluation framework. Instead of focusing solely on response length or fluency, the evaluation emphasizes practical decision-support quality using the following criteria:

- Business understanding
- Numerical accuracy
- Evidence-based reasoning
- Hallucination resistance
- Handling of incomplete information
- Executive readability
- Actionable recommendation quality

The objective is to identify prompting strategies that produce reliable, transparent, and business-oriented recommendations while minimizing unsupported assumptions.
The overall workflow of the proposed AI-assisted decision-support framework is illustrated in **Figure 1**.

**Figure 1. Proposed AI-Assisted Enterprise Decision Support Framework**



## IV. METHODOLOGY

This research presents a reproducible methodology for evaluating the effectiveness of prompt engineering strategies in AI-assisted enterprise decision support. The proposed workflow integrates Python-based exploratory data analysis (EDA) with Large Language Models (LLMs) to generate and evaluate business recommendations. The methodology consists of six major stages: dataset acquisition, data preprocessing, exploratory data analysis, business summary generation, prompt engineering experiments, and comparative evaluation. The overall research workflow is illustrated in **Figure 1**.

### A. Dataset Description

The experiments were conducted using the **Tableau Sample Superstore** dataset, a publicly available retail sales dataset widely used for business analytics, visualization, and machine learning research. The dataset represents transactional information from a fictional retail company and contains historical sales records collected over multiple years.

The dataset consists of **9,994 transaction records** and **21 business attributes**, providing sufficient information for evaluating enterprise business performance. The attributes include customer information, product details, geographical regions, sales, profit, discounts, shipping methods, and order information.

The primary variables used in this research include:

- Order Date
- Ship Date
- Customer Segment
- Region
- State
- Product Category
- Sub-Category
- Sales
- Profit
- Discount
- Quantity

The dataset was selected because it closely resembles real-world enterprise sales data and provides multiple business dimensions that support strategic decision-making.

### B. Data Preprocessing

Before performing the analysis, the dataset underwent preprocessing to ensure data quality and consistency. Data preprocessing was implemented using **Python** with the **Pandas** library.

The preprocessing stage involved the following tasks:

- Loading the dataset into a Pandas DataFrame.
- Verifying the dataset dimensions (9,994 rows × 21 columns).
- Inspecting data types for all attributes.
- Identifying missing values.
- Converting the **Order Date** and **Ship Date** columns into datetime format.
- Verifying numerical attributes including Sales, Profit, Quantity, and Discount.
- Checking overall dataset integrity before analysis.

The missing value analysis indicated that only the **Postal Code** attribute contained a small number of missing records, while all remaining business variables were complete. Since Postal Code was not required for the analytical objectives of this research, the missing values did not significantly affect subsequent analysis.

### C. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the business characteristics of the dataset and identify key performance indicators before interacting with the Large Language Model.

The analysis was conducted using the **Pandas**, **NumPy**, and **Matplotlib** libraries in Python.

The following business analyses were performed:

**1. Descriptive Statistics**

Summary statistics including mean, median, standard deviation, minimum values, and maximum values were generated for numerical business variables such as Sales, Profit, Quantity, and Discount.

**2. Sales Performance Analysis**

Overall business performance was evaluated by computing:

- Total Sales
- Total Profit
- Average Discount

**3. Regional Analysis**

Sales and profit were aggregated across the four geographical regions:

- West
- East
- Central
- South

A bar chart illustrating regional sales distribution was generated using Matplotlib (Figure 2).

**4. Product Category Analysis**

Sales and profit were summarized for the three major product categories:

- Technology
- Furniture
- Office Supplies

This analysis identified Technology as the highest revenue-generating category.

**5. Customer Segment Analysis**

Business performance was analyzed across customer segments to understand purchasing behavior and segment contribution to total sales and profit.

The results of the EDA formed the foundation for generating structured business summaries used in prompt engineering experiments.

**D. Business Summary Generation**

Following exploratory analysis, a structured business summary was automatically generated using Python. The summary consolidated the key business metrics extracted from the dataset into a concise, standardized format.

The business summary included:

- Total Sales
- Total Profit
- Average Discount
- Top-performing States
- Sales by Product Category
- Profit by Region
- Customer Segment Performance

The summary generated served as the **single source of input** for all prompting experiments. By providing identical business information to the Large Language Model in every experiment, differences in generated responses could be attributed solely to the prompting strategy rather than changes in the underlying data.

**E. Prompt Engineering Experiments**

To investigate the influence of prompt design on AI-generated business recommendations, four prompting strategies were evaluated using **Google Gemini Flash**. Each prompting strategy received an identical business summary while differing only in prompt structure.

**1. Zero-Shot Prompting:** The model received only task instructions without examples or predefined reasoning steps. This experiment evaluated the model's ability to independently analyze business information and generate recommendations.

**2. Structured Prompting:** The model was instructed to perform business analysis through predefined analytical stages, including business strengths, weaknesses, supporting evidence, recommendations, and limitations. This approach encouraged systematic reasoning and evidence-based responses.

**3. Role-Based Prompting:** The model was assigned the role of a **Chief Strategy Officer (CSO)** responsible for advising executive leadership. The objective was to evaluate whether assigning professional context improved strategic reasoning and executive-level communication.

**4. Few Shot Prompting:** An example business analysis was provided before the actual task. This prompting strategy demonstrated the expected reasoning style and report structure, enabling the model to generate more consistent and structured business recommendations.

All four experiments were conducted independently using the same business summary to ensure experimental consistency.

**F. Evaluation Criteria**

The responses generated by the Large Language Model were evaluated using qualitative and quantitative criteria commonly applied in enterprise decision-support research. The objective was to assess not only the correctness of the recommendations generated but also their practical usefulness for business decision-making.

The evaluation considered the following criteria:

**1. Business Understanding:** The ability of the model to correctly interpret business performance, identify trends, and recognize strategic issues.

**2. Numerical Accuracy:** The extent to which the recommendations generated correctly referenced numerical values obtained from the dataset without introducing unsupported calculations.

**3. Evidence-Based Reasoning:** The degree to which recommendations were supported by information derived from the business summary rather than unsupported assumptions.

**4. Hallucination Resistance:** The model's ability to explicitly acknowledge insufficient information and avoid generating fabricated or unverifiable conclusions.

**5. Executive Readability:** The clarity, organization, professionalism, and suitability of the generated report for executive decision-makers.

**6. Decision Quality:** The practical usefulness, relevance, and strategic value of the recommendations generated in supporting enterprise decision-making.

Each prompting strategy was comparatively assessed using these evaluation criteria to determine its effectiveness for AI-assisted enterprise decision support.

**G. Software and Development Environment**

The experimental workflow was implemented using the following software environment:

| Component | Specification |
| --- | --- |
| Programming Language | Python 3.x |
| Development Platform | Google Colab |
| Data Analysis Libraries | Pandas, NumPy |
| Visualization Library | Matplotlib |
| Spreadsheet Reader | OpenPyXL |
| Dataset | Tableau Sample Superstore |
| Large Language Model | Google Gemini Flash |
| Version Control | GitHub |

## V. EXPERIMENTAL RESULTS

This section presents the experimental results obtained from the proposed AI-assisted enterprise decision support framework. The evaluation consists of Python-based exploratory data analysis (EDA), business performance analysis, and a comparative assessment of four prompt engineering strategies using Google Gemini Flash. All prompting experiments were conducted using the identical business summary generated from the Tableau Sample Superstore dataset, ensuring that differences in the generated responses resulted solely from variations in prompt design.

### A. Dataset Statistics

The experiments were conducted using the Tableau Sample Superstore dataset containing **9,994 retail transactions** and **21 business attributes**. The dataset includes transactional information such as customer details, product categories, geographical regions, sales, profit, quantity, discounts, shipping methods, and order information.

Python-based preprocessing confirmed that the dataset was suitable for business analytics. Missing value analysis indicated that only the **Postal Code** attribute contained a small number of missing values (11 records), while all remaining variables were complete. Since Postal Code was not required for the analytical objectives of this research, the dataset was retained without additional imputation.

The descriptive statistical analysis generated summary measures including minimum, maximum, mean, median, and standard deviation for all numerical variables, providing an overview of the business performance before further analysis.

### B. Python-Based Business Analysis

Python was used to compute descriptive statistics and generate business performance indicators. The analysis produced the following key performance metrics:

**B. Python-Based Business Analysis**

| Business Metric | Value |
|---|---|
| Total Sales | $2,297,200.86 |
| Total Profit | $286,397.02 |
| Average Discount | 0.16 |
| Total Records | 9,994 |
| Total Variables | 21 |

The generated business summary served as the standardized input for all prompt engineering experiments.

### C. Sales Performance Analysis

Sales performance was evaluated across geographical regions and product categories to identify major revenue contributors.

The total sales amounted to **$2,297,200.86**, indicating a substantial retail transaction volume across multiple product categories and customer segments.

State-level analysis identified **California** as the highest-performing state, followed by **New York**, **Texas**, **Washington**, and **Pennsylvania**. Together, these five states contributed more than half of the organization's total sales, indicating a concentration of business activity within a limited number of geographic markets.

Technology generated the highest product-category sales, followed by Furniture and Office Supplies, demonstrating a relatively balanced distribution of organizational revenue across the three primary product categories.

### D. Profitability Analysis

Overall organizational profit was calculated as **$286,397.02**. Regional profit analysis indicated that the **West** region generated the highest overall profitability, followed closely by the **East** region. In contrast, the **Central** and **South** regions contributed considerably lower profits despite maintaining significant sales volumes.

This finding suggests that sales volume alone does not necessarily translate into higher profitability and highlights the importance of evaluating operational efficiency in addition to revenue generation.

### E. Product Category Analysis

Sales performance across product categories demonstrated that:

- Technology generated the highest revenue.
- Furniture represented the second largest contributor.
- Office Supplies generated slightly lower sales while remaining a significant component of overall business revenue.

The relatively balanced distribution across product categories suggests that organizational revenue is not overly dependent on a single product line, thereby reducing business risk associated with product concentration.

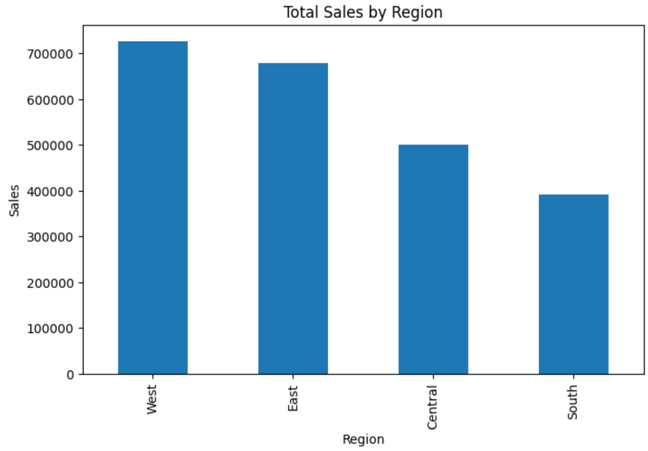
### F. Regional Analysis

Regional sales analysis revealed noticeable differences in market performance.

The **West** region recorded the highest sales, followed closely by the **East** region. The **Central** region demonstrated moderate sales performance, whereas the **South** region generated the lowest total sales among the four geographical regions.

Figure 2 illustrates the regional sales distribution generated using Python and Matplotlib.

**Figure 2. Total Sales by Region**




The visualization clearly demonstrates that the West and East regions dominate organizational sales performance, while the South region represents the smallest revenue contributor.

**G. Customer Segment Analysis**

Customer segment analysis was conducted by aggregating sales and profit across the three primary customer groups:

- Consumer
- Corporate
- Home Office

The analysis revealed meaningful differences in purchasing behavior and revenue contribution across customer segments. These findings provide additional context for understanding customer diversity within the retail business. Although customer segments contributed differently to overall sales performance, further segmentation based on customer profitability, purchasing frequency, and lifetime value was beyond the scope of the current dataset.

**H. Prompt Engineering Experiments**

The generated business summary was provided to Google Gemini Flash using four different prompting strategies. Each experiment used identical business information while varying only the prompt design.

**Experiment 1: Zero-Shot Prompting**

**Prompt**

*You are a Senior Business Strategy Consultant and Data Analytics Expert.*

*Your task is to analyze the business performance summary below and provide evidence-based recommendations for executive decision-making.*

*Instructions:*
*- Base every conclusion ONLY on the information provided.*
*- Do not assume or invent information that is not present.*
*- Clearly distinguish observations from recommendations.*
*- Explain your reasoning before giving recommendations.*
*- Keep the response professional and suitable for a business executive.*

*Provide your analysis using the following structure:*

*## Executive Summary*
*Provide a concise overview of the company's current business performance.*

*## Business Strengths*
*Identify the three strongest aspects of the business based on the data.*

*## Business Risks*
*Identify the three most significant business risks or concerns supported by the data.*

*## Sales Performance Analysis*
*Analyze sales performance across:*
*- Regions*
*- Product Categories*
*- Customer Segments*
*Identify any noticeable trends or opportunities.*

*## Profitability Analysis*
*Evaluate overall profitability and identify factors that may negatively impact profit.*

*## Strategic Recommendations*
*Recommend at least five actionable strategies that management should implement to:*
*- Increase revenue*
*- Improve profitability*
*- Reduce operational risks*
*- Optimize sales performance*

*For each recommendation include:*
*- Business objective*
*- Supporting evidence from the data*
*- Expected business impact*

*## Limitations*
*Explain what additional data would improve the quality of these recommendations.*

*Before making any recommendation, determine whether the available data is sufficient to support the recommendation.*

*If there is insufficient evidence, explicitly state:*
*"Insufficient evidence to draw a reliable conclusion."*

*Do not speculate or fabricate missing information.*
*BUSINESS SUMMARY*
*====================================*
*========================*
*Total Sales: $2,297,200.86*
*Total Profit: $286,397.02*

*Top 5 States by Sales*
*State*
*California 457687.6315*
*New York 310876.2710*
*Texas 170188.0458*
*Washington 138641.2700*
*Pennsylvania 116511.9140*
*Name: Sales, dtype: float64*

*Top 5 Product Categories*
*Category*
*Technology 836154.0330*
*Furniture 741999.7953*
*Office Supplies 719047.0320*
*Name: Sales, dtype: float64*

*Profit by Region*
*Region*
*West 108418.4489*
*East 91522.7800*
*South 46749.4303*
*Central 39706.3625*
*Name: Profit, dtype: float64*

**Output**

***Executive Summary***
*The business generated **$2,297,200.86** in Total Sales and achieved a Total Profit of **$286,397.02**, representing an overall net profit margin of approximately **12.47%**. Revenue is distributed across three main product categories: **Technology** ($836,154.03), **Furniture** ($741,999.80), and **Office Supplies** ($719,047.03).*
*Geographically, revenue generation is concentrated in key states, led by **California** ($457,687.63) and **New York** ($310,876.27). Regional profitability is strongly skewed toward the **West** ($108,418.45) and **East** ($91,522.78) regions, which together contribute over 69% of overall company profits. Conversely, the Central and South regions lag significantly in profit contribution.*

***Business Strengths***
1. ***Balanced Product Revenue Base:** Revenue is evenly spread across three primary product categories, preventing over-reliance on a single line:*
   - ***Technology:** $836,154.03 (36.40% of total sales)*
   - ***Furniture:** $741,999.80 (32.30% of total sales)*
   - ***Office Supplies:** $719,047.03 (31.30% of total sales)*
2. ***High Regional Profit Pillars:** The **West** ($108,418.45) and **East** ($91,522.78) regions perform exceptionally well, together accounting for **$199,941.23** (69.81%) of total business profit.*
3. ***Strong Anchor Markets: California** ($457,687.63) and **New York** ($310,876.27) serve as major sales drivers, combined generating **$768,563.90**, or **33.46%** of total revenue.*

***Business Risks***
1. ***High Geographic Sales Concentration:** The top 5 states—California, New York, Texas, Washington, and Pennsylvania—account for **$1,193,905.13 (51.97%)** of total sales. A localized economic downturn or disruption in these key states poses a risk to top-line performance.*
2. ***Underperforming Regional Profitability:** Profitability in the **Central** ($39,706.36) and **South** ($46,749.43) regions combined represents only **30.19%** of total profits. Lower profit realization in these regions suggests operational inefficiencies or higher cost structures.*
3. ***Unmapped Category Profitability:** While Technology, Furniture, and Office Supplies generate massive sales volume ($2.29M+), category-specific profit figures are unrecorded in the summary. High-volume categories like Furniture or Office Supplies could potentially operate on thin or negative margins without visibility.*

***Sales Performance Analysis***
***Sales by Region***
- ***Top Sales States:** California leads state revenue at **$457,687.63**, followed by New York (**$310,876.27**), Texas (**$170,188.05**), Washington (**$138,641.27**), and Pennsylvania (**$116,511.91**).*
- *Note: Regional aggregate sales totals were not provided in the dataset; regional data is limited strictly to regional profit.*

***Sales by Product Category***

| *Category* | *Total Sales ($)* | *Revenue Share (%)* |
|---|---|---|
| ***Technology*** | *$836,154.03* | *36.40%* |
| ***Furniture*** | *$741,999.80* | *32.30%* |
| ***Office Supplies*** | *$719,047.03* | *31.30%* |
| ***Total*** | ***$2,297,200.86*** | ***100.00%*** |

- ***Observation:** Technology is the primary revenue engine, outperforming Office Supplies by $117,107.00.*

***Sales by Customer Segments***

***Insufficient evidence to draw a reliable conclusion.***
*(No customer segment data, metrics, or breakdowns were provided in the business summary.)*

***Trends & Opportunities***

- ***State Market Expansion:*** *Mid-tier revenue states like Texas ($170.2K), Washington ($138.6K), and Pennsylvania ($116.5K) represent immediate scaling opportunities to reduce dependence on California and New York.*

***Profitability Analysis***

*The business achieved a Total Profit of* ***$286,397.02*** *against* ***$2,297,200.86*** *in Total Sales (Net Margin:* ***~12.47%****).*

***Profit Distribution by Region***

*West*
*$108,418.45 (37.86%)*
*East*
*$91,522.78 (31.96%)*
*South*
*$46,749.43 (16.32%)*
*Central*
*$39,706.36 (13.86%)*

***Factors Affecting Profitability***

- ***Regional Disparity:*** *The West region generates over* ***2.73x*** *the profit of the Central region. This variance indicates higher regional operational costs, heavier discounting, or adverse product mix ratios in the Central and South territories.*
- ***Uncertain Cost/Margin Dynamics:*** *Without unit cost, discounting, or cost-of-goods-sold (COGS) figures, exact drivers of margin compression in low-profit regions cannot be conclusively calculated from the summary alone.*

***Strategic Recommendations***

***Reasoning & Strategy***

*Analysis shows that* ***69.81%*** *of total profits originate from the West and East regions, while the top 5 states generate* ***51.97%*** *of top-line revenue. Meanwhile, Central and South lag severely in profit contribution. To maximize bottom-line performance, executive management must double down on core revenue hubs, audit underperforming regions, and prioritize inventory allocation toward higher-value product categories.*

***1. Capitalize on Core State Anchors (California & New York)***

- ***Business Objective:*** *Increase overall sales revenue by deepening penetration in established high-converting markets.*
- ***Supporting Evidence:*** *California ($457,687.63) and New York ($310,876.27) generated* ***$768,563.90*** *combined (33.46% of total revenue).*
- ***Expected Business Impact:*** *Accelerated top-line growth with lower customer acquisition costs due to existing market strength.*

***2. Operational & Margin Audit in Central and South Regions***

- ***Business Objective:*** *Improve profitability in underperforming territories.*
- ***Supporting Evidence:*** *Central ($39,706.36) and South ($46,749.43) together contribute only* ***$86,455.79*** *(30.19%) of overall profit, despite containing large sales markets such as Texas ($170,188.05).*
- ***Expected Business Impact:*** *Identification of cost leaks, discount abuses, or logistical inefficiencies, leading to expanded regional net margins.*

***3. Inventory and Sales Prioritization for Technology Products***

- ***Business Objective:*** *Optimize sales performance by expanding allocation to top revenue-generating lines.*
- ***Supporting Evidence:*** *Technology leads all categories in revenue at* ***$836,154.03*** *(36.40% of sales).*
- ***Expected Business Impact:*** *Maximized average order value and increased revenue throughput across core sales channels.*

***4. Revenue Diversification in Mid-Tier States***

- ***Business Objective:*** *Reduce operational risk tied to geographic sales concentration.*
- ***Supporting Evidence:*** *Top 5 states make up* ***51.97%*** *($1.19M) of sales. Texas ($170.2K), Washington ($138.6K), and Pennsylvania ($116.5K) are proven secondary revenue pillars.*
- ***Expected Business Impact:*** *Protection against localized economic shocks by establishing a broader revenue foundation.*

***5. Category-Level Profit Tracking & Governance***

- ***Business Objective:*** *Protect overall profit margins against potential category-level losses.*
- ***Supporting Evidence:*** *Non-Technology categories (Furniture and Office Supplies) account for* ***$1,461,046.83*** *(63.60% of total revenue), but category-specific profits are untracked in the dataset.*
- ***Expected Business Impact:*** *Elimination of low-margin or loss-leading products within Furniture and Office Supplies to protect overall company profitability (~12.47%).*

***Limitations***

*To provide a more comprehensive, granular analysis and higher-confidence strategic recommendations, the following data points would be required:*

1. ***Customer Segment Data:*** *Metrics detailing B2B vs. Consumer segments, customer acquisition costs (CAC), lifetime value (LTV), and transaction volumes.*
2. ***Category & State Profitability:*** *Profit breakdowns by product category and individual state to determine exact product-level margins.*
3. ***Regional Sales Totals:*** *Aggregate sales revenue figures for West, East, Central, and South regions (only state-level sales and regional profits were provided).*
4. ***Time-Series / Temporal Data:*** *Transaction dates, monthly/quarterly historical figures, or yearly trends to assess growth rates and seasonality.*
5. ***Cost & Pricing Detail:*** *Operational expenses, Cost of Goods Sold (COGS), shipping fees, and discount rates per region/category.*

**Discussion**

Zero-Shot Prompting produced comprehensive business recommendations with strong executive readability. The model correctly identified major business trends and generated actionable recommendations using the available numerical information. However, several observations extended beyond the explicitly available evidence, indicating occasional reliance on inferred reasoning. Although the report generated was informative and well organized, unsupported assumptions reduced its overall factual reliability when compared with more structured prompting strategies.

**Experiment 2: Structured Prompting**

**Prompt**

*You are an AI Business Analyst.*

*Perform the analysis using the following process.*

*Step 1:*
*Summarize the business.*

*Step 2:*
*Identify business strengths.*

*Step 3:*
*Identify business weaknesses.*

*Step 4:*
*Support every observation with numerical evidence.*

*Step 5:*
*If the data is insufficient,*
*reply:*

*"Insufficient evidence."*

*Never make assumptions.*

*Finally,*

*Provide five recommendations ranked by expected business impact.*

*Business Summary:*

*=============================================================*
*BUSINESS SUMMARY*
*=============================================================*
*Total Sales: $2,297,200.86*
*Total Profit: $286,397.02*

*Top 5 States by Sales*
*State*
*California      457687.6315*
*New York        310876.2710*
*Texas           170188.0458*
*Washington      138641.2700*
*Pennsylvania    116511.9140*
*Name: Sales, dtype: float64*

*Top 5 Product Categories*
*Category*
*Technology        836154.0330*
*Furniture         741999.7953*
*Office Supplies   719047.0320*
*Name: Sales, dtype: float64*

*Profit by Region*
*Region*
*West       108418.4489*
*East       91522.7800*
*South      46749.4303*
*Central    39706.3625*
*Name: Profit, dtype: float64*

**Output**

***Step 1: Business Summary***

- ***Total Sales:*** *$2,297,200.86*
- ***Total Profit:*** *$286,397.02 (Overall net profit margin: ~12.47%)*
- ***Top States by Sales:*** *California ($457,687.63), New York ($310,876.27), Texas ($170,188.05), Washington ($138,641.27), and Pennsylvania ($116,511.91).*
- ***Product Categories Listed:*** *Technology ($836,154.03), Furniture ($741,999.80), and Office Supplies ($719,047.03).*

- ***Profit by Region:*** *West ($108,418.45), East ($91,522.78), South ($46,749.43), and Central ($39,706.36).*

***Step 2: Business Strengths***

1. ***High Profit Contribution from West & East Regions:*** *The West region generates $108,418.45 (37.86% of total profit) and the East region generates $91,522.78 (31.96% of total profit), making up 69.82% of all business profit.*
2. ***Strong Revenue Concentration in Key States:*** *California ($457,687.63) and New York ($310,876.27) are the leading revenue drivers, generating a combined $768,563.90 (33.46% of total sales).*
3. ***Primary Sales Category:*** *Technology is the highest performing category by revenue, contributing $836,154.03 (36.40% of total sales).*

***Step 3: Business Weaknesses***

1. ***Underperforming Regional Profitability:*** *The Central region ($39,706.36; 13.86% of total profit) and South region ($46,749.43; 16.32% of total profit) lag significantly behind the West and East regions in profit generation.*
2. ***Sales Disparity Among Top States:*** *Sales drop significantly after the top two states; 5th ranked Pennsylvania ($116,511.91) generates roughly a quarter of California's sales volume ($457,687.63).*

***Step 4: Numerical Evidence***

***Regional Profit Breakdown***

| ***Region*** | ***Profit ($)*** | ***Share of Total Profit (%)*** |
|---|---|---|
| ***West*** | *$108,418.45* | *37.86%* |
| ***East*** | *$91,522.78* | *31.96%* |
| ***South*** | *$46,749.43* | *16.32%* |
| ***Central*** | *$39,706.36* | *13.86%* |
| ***Total*** | ***$286,397.02*** | ***100.00%*** |

***Top 5 States Sales Breakdown***

| ***State*** | ***Sales ($)*** | ***Share of Total Sales (%)*** |
|---|---|---|
| ***California*** | *$457,687.63* | *19.92%* |
| ***New York*** | *$310,876.27* | *13.53%* |
| ***Texas*** | *$170,188.05* | *7.41%* |
| ***Washington*** | *$138,641.27* | *6.03%* |
| ***Pennsylvania*** | *$116,511.91* | *5.07%* |
| ***Top 5 Total*** | ***$1,193,905.13*** | ***51.97%*** |

***Product Category Sales Breakdown***

| ***Category*** | ***Sales ($)*** | ***Share of Total Sales (%)*** |
|---|---|---|
| ***Technology*** | *$836,154.03* | *36.40%* |
| ***Furniture*** | *$741,999.80* | *32.30%* |
| ***Office Supplies*** | *$719,047.03* | *31.30%* |
| ***Total*** | ***$2,297,200.86*** | ***100.00%*** |

***Step 5: Data Sufficiency Evaluation***

*"Insufficient evidence."*

***Key Data Missing:***

- ***No Profit Figures by Category or State:*** *Sales are provided for states and categories, but profit is provided only by region. Profit margins per category or per state cannot be calculated.*
- ***No Sales Figures by Region:*** *Profit is provided by region, but regional sales are missing. Profit margins per region (e.g., Central profit vs. Central sales) cannot be calculated.*
- ***Incomplete Category List:*** *The data header indicates "Top 5 Product Categories", but only 3 categories are listed.*
- ***No Cost, Operational, or Time-Series Metrics:*** *There is no information regarding COGS, discounting, volume, operational expense, or trend over time.*

*Because zero assumptions are permitted, it is impossible to quantify or rank five evidence-backed business recommendations by expected impact without making unverified assumptions about category profitability, regional sales baselines, or underlying cost drivers.*

**Discussion**

Structured Prompting generated the most systematic and evidence-based business analysis. The predefined reasoning structure encouraged the model to separate strengths, weaknesses, supporting evidence, recommendations, and limitations into clearly defined sections. Importantly, the model consistently acknowledged situations where insufficient information was available instead of generating unsupported conclusions. This approach produced highly transparent and reproducible outputs suitable for enterprise decision support.

**Experiment 3: Role-Based Prompting**

**Prompt**

*You are the Chief Strategy Officer (CSO) of a Fortune 500 retail company.*

*You have been asked by the CEO to review the company's sales performance.*

*Use only the provided business summary.*

*Do not speculate.*

*Provide:*

*Executive Summary*

*Financial Performance*

*Operational Risks*

*Growth Opportunities*

*Top Five Executive Decisions*

*Expected Business Impact*

*Business Summary:*

*You are the Chief Strategy Officer (CSO) of a Fortune 500 retail company.*

*You have been asked by the CEO to review the company's sales performance.*

*Use only the provided business summary.*

*Do not speculate.*

*Provide:*

*Executive Summary*

*Financial Performance*

*Operational Risks*

*Growth Opportunities*

*Top Five Executive Decisions*

*Expected Business Impact*

*BUSINESS SUMMARY*

*=============================================================*

*Total Sales: $2,297,200.86*
*Total Profit: $286,397.02*

*Top 5 States by Sales*
*State*
*California      457687.6315*
*New York        310876.2710*
*Texas           170188.0458*
*Washington      138641.2700*
*Pennsylvania    116511.9140*
*Name: Sales, dtype: float64*

*Top 5 Product Categories*
*Category*
*Technology         836154.0330*
*Furniture          741999.7953*
*Office Supplies    719047.0320*
*Name: Sales, dtype: float64*

*Profit by Region*
*Region*
*West       108418.4489*
*East        91522.7800*
*South       46749.4303*
*Central     39706.3625*
*Name: Profit, dtype: float64*

**Output**

***MEMORANDUM***

***TO:*** *Chief Executive Officer*

***FROM:*** *Chief Strategy Officer*

***SUBJECT:*** *Executive Sales Performance & Strategic Review*

***Executive Summary***

*Our company achieved **$2,297,200.86** in total sales, generating **$286,397.02** in total profit (a overall profit margin of **12.47%**).*

*Performance is heavily anchored by key geographic strongholds—most notably **California** ($457,687.63) and **New York** ($310,876.27)—and driven by three primary product categories: **Technology**, **Furniture**, and **Office Supplies**. Profitability varies significantly across regions, led by the **West** ($108,418.45) and **East** ($91,522.78), while the **Central** and **South** regions lag behind.*

*Below is the detailed breakdown based strictly on our reported financial data.*

***Financial Performance***

***Overall Revenue & Profitability***

- ***Total Sales:*** *$2,297,200.86*
- ***Total Profit:*** *$286,397.02*
- ***Overall Profitability:*** *12.47%*

***Regional Profit Performance***

*The **West** and **East** regions drive **69.8%** of total company profit, while the **South** and **Central** regions account for the remaining **30.2%**.*

| ***Region*** | ***Total Profit*** | ***Share of Profit*** |
|---|---|---|
| ***West*** | *$108,418.45* | *37.86%* |
| ***East*** | *$91,522.78* | *31.96%* |
| ***South*** | *$46,749.43* | *16.32%* |
| ***Central*** | *$39,706.36* | *13.86%* |

***Sales Concentration by Top 5 States***

*The top 5 states generate **$1,193,905.13**, representing **51.97%** of our total sales.*

| ***State*** | ***Total Sales*** | ***Share of Total Sales*** |
|---|---|---|
| ***California*** | *$457,687.63* | *19.92%* |
| ***New York*** | *$310,876.27* | *13.53%* |
| ***Texas*** | *$170,188.05* | *7.41%* |
| ***Washington*** | *$138,641.27* | *6.03%* |
| ***Pennsylvania*** | *$116,511.91* | *5.07%* |

***Product Category Sales Breakdown***

*The three reported categories account for **100%** ($2,297,200.86) of company sales:*

| ***Category*** | ***Total Sales*** | ***Share of Total Sales*** |
|---|---|---|
| ***Technology*** | *$836,154.03* | *36.40%* |
| ***Furniture*** | *$741,999.80* | *32.30%* |
| ***Office Supplies*** | *$719,047.03* | *31.30%* |

***Operational Risks***

1. ***Geographic Sales Risk:*** *Over half of all sales **(51.97%)** are concentrated in just 5 states, with California and New York alone generating **33.45%** ($768,563.90) of total revenue.*
2. ***Regional Profit Underperformance:*** *The Central ($39,706.36) and South ($46,749.43) regions combined yield less profit than the East region alone ($91,522.78) and less than half of the West region ($108,418.45).*
3. ***Data & Reporting Discrepancy:*** *The performance data header specifies "Top 5 Product Categories", but only 3 product categories exist in the record. Notably, these 3 categories sum exactly to total corporate sales ($2,297,200.86), indicating either a reporting naming error or lack of additional product diversity.*

***Growth Opportunities***

1. ***Capitalizing on Leading Revenue States:*** *Strengthen presence in California ($457,687.63) and New York ($310,876.27), our two largest geographic revenue drivers.*
2. ***Profit Optimization in Central & South:*** *Operational review of the Central ($39,706.36) and South ($46,749.43) regions to elevate their earnings closer to West and East benchmarks.*
3. ***Product Revenue Balance:*** *Expand sales momentum across all three existing product lines—Technology ($836,154.03), Furniture ($741,999.80), and Office Supplies ($719,047.03)—which currently maintain a balanced contribution to overall sales volume.*

***Top Five Executive Decisions***

1. ***Prioritize Resource Allocation in Top 5 States:*** *Direct market development resources to California, New York, Texas, Washington, and Pennsylvania to safeguard 51.97% of total sales.*
2. ***Initiate Regional Performance Reviews:*** *Conduct operational assessments in the Central and South regions to address lower profit contributions ($39,706.36 and $46,749.43, respectively).*
3. ***Focus Supply Chain Focus on Technology:*** *Allocate priority operational capacity to **Technology** ($836,154.03), our largest single sales category.*
4. ***Replicate West & East Regional Practices:*** *Analyze and model the operational practices of the West ($108,418.45) and East ($91,522.78) regions across remaining markets.*
5. ***Standardize Data Reporting Protocols:*** *Audit reporting systems to correct category labeling and ensure exact alignment between data headers and underlying category counts.*

***Expected Business Impact***

- ***Increased Profitability in Lagging Markets:*** *Improving operational performance in the Central and South regions toward West/East averages directly boost total company profit ($286,397.02).*

- ***Revenue Protection:*** *Focused focus on the top 5 states secures the $1,193,905.13 sales base.*
- ***Streamlined Category Management:*** *Aligning resources around Technology, Furniture, and Office Supplies ensures support for 100% of total company revenue.*
- ***Reporting Accuracy:*** *Eliminating discrepancies in financial summaries establishes clear visibility for future executive decision-making.*

**Discussion**

Assigning the model the role of a Chief Strategy Officer resulted in recommendations presented from an executive management perspective. The report emphasized strategic planning, operational priorities, and business growth while maintaining strong readability and professional communication. Although factual accuracy remained high, several recommendations reflected executive reasoning rather than strictly evidence-based analytical conclusions, illustrating the influence of contextual role assignment on LLM behavior.

**Experiment 4: Few - Shot Prompting**

**Prompt**

You are an AI Business Strategy Consultant.

Below is an example of how business analysis should be performed.

-------------------------

Example

Business Summary

Total Sales: $500,000

Total Profit: $60,000

Top Region: West

Weak Region: South

Technology generated the highest revenue.

Example Analysis

Strength: Technology is the strongest-performing category because it contributes the highest revenue.

Weakness: The South region contributes the least profit and should be investigated.

Recommendation: Increase investment in Technology while reviewing operational performance in the South region.

The recommendation is supported by the available data.

-------------------------

Now perform the same style of analysis for the following business summary.

Instructions:

- Follow the same analytical style as the example.
- Base every conclusion ONLY on the provided data.
- Never invent information.
- If information is missing, explicitly write:
  "Insufficient evidence."
- Keep the response concise, evidence-based, and executive-friendly.

Provide:

1. Executive Summary

2. Business Strengths

3. Business Weaknesses

4. Supporting Numerical Evidence

5. Five Executive Recommendations

6. Data Limitations

```
BUSINESS SUMMARY
============================================================
Total Sales: $2,297,200.86
Total Profit: $286,397.02

Top 5 States by Sales
State
California      457687.6315
New York        310876.2710
Texas           170188.0458
Washington      138641.2700
Pennsylvania    116511.9140
Name: Sales, dtype: float64

Top 5 Product Categories
Category
Technology         836154.0330
Furniture          741999.7953
Office Supplies    719047.0320
Name: Sales, dtype: float64

Profit by Region
Region
West       108418.4489
```

East 91522.7800
South 46749.4303
Central 39706.3625
Name: Profit, dtype: float64

**Output**

**1. Executive Summary**

The business generated **$2,297,200.86** in Total Sales and **$286,397.02** in Total Profit. **Technology** represents the top revenue-generating product category, while **California** leads all states in sales. Regionally, the **West** region is the top profit performer, whereas the **Central** and **South** regions contribute the lowest overall profits. Detailed profit metrics by category and individual state sales by region are absent from the summary.

**2. Business Strengths**

- **Top Revenue Category:** Technology is the strongest sales driver among reported categories ($836,154.03).
- **Key Regional Profit Driver:** The West region generates the highest profit ($108,418.45), closely followed by the East region ($91,522.78).
- **Core State Markets:** California ($457,687.63) and New York ($310,876.27) lead state-level sales performance by a significant margin.

**3. Business Weaknesses**

- **Lowest Regional Profitability:** The Central region contributes the lowest profit across all regions ($39,706.36).
- **Underperforming South Region:** The South region represents the second-lowest profit contributor ($46,749.43).
- **Category Profitability Unknown:** Insufficient evidence to determine whether top sales categories (like Technology or Furniture) are actually profitable.

**4. Supporting Numerical Evidence**

| Metric Category | Subject / Entity | Value |
|---|---|---|
| **Totals** | Total Sales | $2,297,200.86 |
| | Total Profit | $286,397.02 |
| **Top States (Sales)** | 1. California | $457,687.63 |
| | 2. New York | $310,876.27 |
| | 3. Texas | $170,188.05 |
| | 4. Washington | $138,641.27 |
| | 5. Pennsylvania | $116,511.91 |
| **Product Categories (Sales)** | 1. Technology | $836,154.03 |
| | 2. Furniture | $741,999.80 |
| | 3. Office Supplies | $719,047.03 |
| **Regions (Profit)** | 1. West | $108,418.45 |
| | 2. East | $91,522.78 |
| | 3. South | $46,749.43 |
| | 4. Central | $39,706.36 |

**5. Five Executive Recommendations**

1. **Invest in High-Performing Regions:** Allocate growth capital to the West ($108,418.45 profit) and East ($91,522.78 profit) regions to leverage existing profit momentum.
2. **Review Central & South Operations:** Conduct an operational performance audit in the Central ($39,706.36 profit) and South ($46,749.43 profit) regions to identify drivers of low profitability.
3. **Protect Core Revenue States:** Maintain focused sales efforts in California ($457,687.63) and New York ($310,876.27), which collectively generate over $768,000 in sales.
4. **Capitalize on Technology Demand:** Prioritize inventory and promotional support for Technology ($836,154.03), the company's highest-grossing category.
5. **Commission Granular Profit Reporting:** Require detailed profit data by product category and by state before making major capital reallocation decisions; further specific margin optimization currently has **insufficient evidence**.

**6. Data Limitations**

- **Missing Product Categories:** Despite being labeled "Top 5 Product Categories," only 3 categories are provided in the data.
- **Lack of Category Profitability:** No profit data is provided for Technology, Furniture, or Office Supplies.

- **Lack of State Profitability:** Profit figures are reported only at the regional level, not by individual state.
- **Lack of Regional Sales:** Sales figures are provided for top states and categories but not summarized by region.
- **Missing Operational & Time Metrics:** Insufficient evidence regarding time periods, profit margins (%), order volumes, or operational costs.

**Discussion**

Few-Shot Prompting demonstrated highly consistent reasoning by following the analytical example provided within the prompt. The recommendations generated closely aligned with the expected reporting style while minimizing unsupported assumptions. Responses remained well structured, transparent, and strongly supported by the available business evidence. Among all evaluated prompting strategies, Few-Shot Prompting produced one of the most reliable outputs for enterprise decision support.

**Table 2. Comparative Evaluation of Prompting Strategies**

| Evaluation Criterion | Zero-Shot | Structured | Role-Based | Few-Shot |
|---|---|---|---|---|
| Business Understanding | Excellent | Excellent | Excellent | Excellent |
| Numerical Accuracy | High | Very High | Very High | Very High |
| Evidence-Based Reasoning | High | Excellent | Excellent | Excellent |
| Hallucination Resistance | Good | Excellent | Very Good | Excellent |
| Handling Missing Information | Good | Excellent | Very Good | Excellent |
| Executive Readability | Excellent | Very Good | Excellent | Very Good |
| Actionable Recommendations | Excellent | Very Good | Excellent | Very Good |
| Overall Performance | ★★★★☆ | ★★★★★ | ★★★★★ | ★★★★★ |

**I. Discussion**

The comparative evaluation demonstrates that **prompt engineering significantly influences the quality of AI-generated enterprise decision support**. Although all four prompting strategies successfully interpreted the business summary, noticeable differences emerged in reasoning quality, transparency, and recommendation reliability.

- **Structured Prompting** consistently produced the most evidence-based responses because the predefined analytical workflow encouraged the model to organize observations systematically, support conclusions with numerical evidence, and explicitly acknowledge missing information instead of generating unsupported assumptions.
- **Few-Shot Prompting** achieved comparable performance by leveraging an example analysis to guide reasoning style and response structure. This approach improved consistency across recommendations while maintaining high factual accuracy and minimizing hallucinations.
- **Role-Based Prompting** generated highly readable executive reports and emphasized strategic business communication. Assigning the model the role of a Chief Strategy Officer encouraged practical decision-making; however, some recommendations reflected strategic interpretation beyond the available numerical evidence.
- **Zero-Shot Prompting** produced comprehensive analyses without requiring additional examples or structured guidance. While the recommendations were generally accurate and useful, the model occasionally inferred conclusions not explicitly supported by the business summary, reducing overall transparency compared with the Structured and Few-Shot approaches.

Overall, the experimental findings indicate that carefully designed prompting strategies substantially improve the reliability, transparency, and practical value of AI-generated business recommendations. These results highlight the importance of prompt engineering as a critical component of trustworthy AI-assisted enterprise decision support systems and demonstrate its potential to enhance business analytics in real-world organizational environments.

## VI. CONCLUSION

This research presented an AI-assisted enterprise decision support framework that integrates Python-based business analytics with Large Language Models (LLMs) to generate evidence-based business recommendations. The proposed methodology combined exploratory data analysis (EDA), business summary generation, prompt engineering, and comparative evaluation to investigate how different prompting strategies influence the quality of AI-generated business insights.

The experimental evaluation was conducted using the Tableau Sample Superstore dataset containing 9,994 retail transactions and 21 business attributes. Python libraries, including Pandas, NumPy, and Matplotlib, were used to perform data preprocessing, descriptive statistical analysis, business performance evaluation, and visualization. The resulting business summary was provided to Google Gemini Flash using four prompt engineering strategies: Zero-Shot Prompting, Structured Prompting, Role-Based Prompting, and Few-Shot Prompting.

Comparative analysis demonstrated that prompt engineering significantly influences the quality, transparency, and reliability of AI-generated business recommendations. Although all four prompting strategies successfully interpreted the business summary, noticeable differences were observed in reasoning quality, factual consistency, evidence utilization, and executive readability.

Among the evaluated approaches, **Structured Prompting** consistently generated the most systematic and evidence-based responses by organizing business analysis into predefined reasoning stages while explicitly acknowledging insufficient information when appropriate. **Few-Shot Prompting** produced highly consistent and reliable recommendations by leveraging example-driven reasoning, thereby reducing unsupported assumptions and improving response consistency. **Role-Based Prompting** generated well-structured executive reports that emphasized strategic decision-making and organizational planning, making it particularly suitable for executive communication. In contrast, **Zero-Shot Prompting** produced comprehensive recommendations with minimal guidance but occasionally introduced conclusions that extended beyond the available business evidence.

The findings indicate that carefully designed prompting strategies can substantially improve the effectiveness of Large Language Models in enterprise decision support by enhancing factual accuracy, reducing hallucinations, improving evidence-based reasoning, and generating recommendations that are more suitable for executive decision-making. These results demonstrate that prompt engineering is not merely a method for interacting with LLMs but an essential component for developing trustworthy AI-assisted business intelligence systems.

The proposed framework provides a reproducible workflow that integrates traditional data analytics with Large Language Models for enterprise decision support. By maintaining identical business inputs across multiple prompting strategies and evaluating responses using standardized criteria, this research offers a practical methodology for assessing AI-generated business recommendations in real-world organizational settings. The study contributes to the growing body of research on prompt engineering and highlights its practical significance in improving the transparency, reliability, and decision-making capabilities of AI systems used in modern enterprise analytics.

Overall, this research demonstrates that the combination of Python-based business analytics and carefully engineered prompts enables Large Language Models to generate meaningful, evidence-based business recommendations while supporting responsible and trustworthy AI adoption within enterprise decision support systems.

## VII. FUTURE SCOPE

Although this study demonstrates the effectiveness of prompt engineering for AI-assisted enterprise decision support, several opportunities exist for extending the proposed framework and improving the robustness of AI-generated business recommendations.

One important direction for future research is the **comparative evaluation of multiple Large Language Models**. This study focused on **Google Gemini Flash**; however, future work should compare the performance of other state-of-the-art LLMs, including **OpenAI GPT**, **Anthropic Claude**, and **Meta Llama**. Evaluating multiple models using identical business datasets and prompting strategies would provide deeper insights into differences in reasoning capability, factual accuracy, consistency, hallucination resistance, and executive decision-making quality. Such comparative analyses would help organizations select the most appropriate LLM for specific enterprise applications.

Future studies may also investigate the integration of **Retrieval-Augmented Generation (RAG)** into enterprise decision-support systems. In the current framework, recommendations were generated solely from the summarized business information provided in the prompt. A RAG-based architecture could retrieve relevant enterprise documents, business policies, historical reports, financial records, or domain-specific knowledge before response generation. By grounding the model's responses in external organizational knowledge, RAG has the potential to improve factual accuracy, reduce hallucinations, and generate more context-aware recommendations for business decision-makers.

Another promising research direction is the development of **AI agent-based enterprise decision-support systems**. Instead of relying on a single Large Language Model, autonomous AI agents could collaboratively perform specialized tasks such as data preprocessing, statistical analysis, financial evaluation, risk assessment, recommendation generation, and report validation. Such modular agent-based architectures could improve scalability, automation, and decision reliability while reducing human intervention in complex analytical workflows.

Future research should also investigate **multi-agent AI systems**, where multiple intelligent agents cooperate to solve business problems through coordinated reasoning and verification. For example, one agent may perform business analytics, another may validate numerical calculations, a third may assess financial risk, and a fourth may review the recommendations generated for consistency and completeness. Multi-agent collaboration offers the potential to improve analytical depth, reduce reasoning errors, and enhance the trustworthiness of AI-generated outputs for enterprise environments.

The incorporation of **Explainable Artificial Intelligence (XAI)** represents another significant extension of this work. While the present study evaluated the quality of AI-generated recommendations, future systems should provide transparent explanations describing how conclusions were derived from the underlying business data. Integrating explainability techniques with Large Language Models would improve stakeholder trust, facilitate human oversight, and support regulatory compliance in domains where transparency is essential, such as finance, healthcare, and public administration.

Future investigations may further expand the proposed framework by evaluating additional prompt engineering techniques, including **Chain-of-Thought Prompting**, **Tree-of-Thought Reasoning**, **Self-Consistency Prompting**, and **ReAct (Reasoning and Acting)** frameworks. These advanced prompting approaches may improve analytical reasoning, reduce unsupported conclusions, and enhance decision quality for complex enterprise scenarios.

Another valuable extension involves applying the proposed framework to **multiple real-world datasets** across diverse industries. While this research utilized a retail sales dataset, future studies should evaluate enterprise datasets from healthcare, banking, manufacturing, supply chain management, telecommunications, and insurance to examine the generalizability of the proposed methodology. Cross-domain evaluations would provide stronger evidence regarding the applicability of AI-assisted decision-support systems across different business environments.

Finally, future work should incorporate **human expert evaluation** alongside automated assessment metrics. Business analysts, financial managers, and domain specialists could evaluate AI-generated recommendations based on practical relevance, strategic usefulness, clarity, and decision quality. Combining expert judgment with quantitative evaluation metrics would provide a more comprehensive assessment of Large Language Models in enterprise decision support.

## VIII. REFERENCES


[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention Is All You Need," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[2] T. Brown *et al*., "Language Models are Few-Shot Learners," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 1877–1901, 2020.

[3] J. Wei *et al*., "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, pp. 24824–24837, 2022.

[4] OpenAI, "GPT-4 Technical Report," *arXiv preprint* arXiv:2303.08774, 2023. (arXiv)

[5] R. Anil *et al*., "Gemini: A Family of Highly Capable Multimodal Models," *arXiv preprint* arXiv:2312.11805, 2023. (arXiv)

[6] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[7] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why Should I Trust You? Explaining the Predictions of Any Classifier," *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 1135–1144, 2016.

[8] E. Tabassi, "Artificial Intelligence Risk Management Framework (AI RMF 1.0)," National Institute of Standards and Technology (NIST AI 100-1), 2023. (NIST)

[9] C. Autio, R. Schwartz, J. Dunietz, S. Jain, M. Stanley, E. Tabassi, P. Hall, and K. Roberts, "Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile," National Institute of Standards and Technology (NIST AI 600-1), 2024. (NIST)

[10] T. H. Davenport and J. G. Harris, *Competing on Analytics: The New Science of Winning*. Boston, MA, USA: Harvard Business School Press, 2007.

[11] E. Turban, R. Sharda, and D. Delen, *Decision Support and Business Intelligence Systems*, 10th ed. Pearson, 2015.

[12] S. Alter, *Decision Support Systems: Current Practice and Continuing Challenges*. Addison-Wesley, 1980.

[13] Microsoft, *Microsoft Responsible AI Standard, Version 2*, Microsoft Corporation, Redmond, WA, USA, 2022.

[14] D. Kahneman, O. Sibony, and C. R. Sunstein, *Noise: A Flaw in Human Judgment*. New York, NY, USA: Little, Brown Spark, 2021.

[15] T. Mitchell, *Machine Learning*. New York, NY, USA: McGraw-Hill, 1997.

[16] F. Chollet, *Deep Learning with Python*, 2nd ed. Shelter Island, NY, USA: Manning Publications, 2021.

[17] C. C. Aggarwal, *Neural Networks and Deep Learning*. Cham, Switzerland: Springer, 2018.

[18] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Hoboken, NJ, USA: Pearson, 2021.

[19] Tableau Software, "Sample Superstore Dataset," Tableau Public Sample Data, Available: https://www.tableau.com/learn/sample-data

[20] Google DeepMind, "Gemini Technical Report and Model Family Documentation," Google DeepMind, 2024. (arXiv)